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Effects of urea infusion and ruminal degradable protein concentration on microbial growth, digestibility, and fermentation in continuous culture¹

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ABSTRACT: The effects of urea and rumen-degradable protein (RDP) on microbial growth, digestibility, and fermentation were examined using dual-flow continuous culture. The experimental design was a 4 × 4 Latin square with a 2 × 2 factorial arrangement of treatments. Factors were urea infusion (0.4 g/L of artificial saliva) and RDP concentration, and the treatments were as follows: 1) low RDP (8% of dietary dry matter) without urea (LDNU), 2) high RDP (11% of dietary dry matter) without urea (HDNU), 3) low RDP (8% of dietary dry matter) with urea (LDU), and 4) high RDP (11% of dietary dry matter) with urea (HDU). The LDNU (i.e., negative control) and HDNU treatments were formulated to be nitrogen limiting. Results indicated that infusion of urea increased all digestibility measurements ($P < 0.05$), which in turn increased ($P < 0.05$) volatile fatty acid, NH_3 nitrogen, trichloroacetic acid-soluble nitrogen, and soluble protein concentrations. Increasing dietary RDP improved dry matter and organic matter digestibility ($P < 0.05$) but did not alter

acid detergent fiber or nonfiber carbohydrate digestibilities ($P > 0.05$). Isobutyrate concentration decreased ($P = 0.05$) with increased RDP. Increased dietary RDP increased crude protein degradation and soluble protein concentration ($P < 0.05$), but NH_3 nitrogen, trichloroacetic acid-soluble nitrogen, and peptide nitrogen were unaffected by changing RDP levels. Microbial growth efficiency was 19.9, 24.9, 28.0, and 32.2 g N/g organic matter truly digested for LDNU, HDNU, LDU, and HDU, respectively, and was significantly improved both by urea infusion ($P = 0.002$) and increased RDP concentration ($P = 0.021$). The interactions of urea and RDP ($P < 0.05$) were explained by the high digestibility of neutral detergent fiber, nonstructural carbohydrate, and especially hemicellulose, with the HDNU treatment. The results of this study indicated that hemicellulose-degrading bacteria were able to effectively compete with nonstructural carbohydrate-degrading bacteria for available peptide and amino acid nitrogen. Further, the extent of protein degradation was dependent on the availability of NH_3 nitrogen in the system.

Key Words: Hemicelluloses, Microbial Protein Synthesis, Protein Degradation, Rumen, Urea

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Introduction

Rumen microbial growth is dependent on the availability of N in the form of peptides, AA, and NH_3 (Russell et al., 1992). To meet the N demands of rumen microbes, the NRC (1989) recommended 60 to 65% of CP as ruminally degradable protein (RDP), and roughly 50% of the

RDP as soluble protein. However, a significant proportion of RDP may flow out of the rumen with the liquid and solid phases (McCarthy et al., 1989), reducing its availability for utilization by ruminal microbes.

Russell et al. (1992) indicated that nonstructural carbohydrate (NSC)-degrading bacteria are the primary users of peptide and AA nitrogen, whereas structural carbohydrate-degrading bacteria only utilize NH_3 . Jones et al. (1998) reported that, with diets containing >40% NSC, peptide and AA nitrogen addition above the requirement needed by NSC-degrading bacteria would not be converted to NH_3 , in turn reducing NH_3 -N below 5 mg/dL, which is regarded to limit microbial growth (Satter and Slyter, 1974).

Accordingly, two scenarios could arise for cattle consuming large amounts of NSC and RDP: 1) fast ruminal passage rate decreases ruminal peptide, AA, and NH_3 concentrations and increases ruminally undegradable protein (RUP) flow; or 2) peptides are rapidly metabo-

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Table 1. Composition of diets and calculated protein contribution of individual dietary ingredients

Ingredient	Low degradable		High degradable		Inoculum ^a	
	Percentage of DM	Theoretical protein contribution, % of CP ^b	Percentage of DM	Theoretical protein contribution, % of CP	Percentage of DM	Theoretical protein contribution, % of CP
Grass hay	22.3	6.21	21.9	6.28	22.1	6.46
Alfalfa hay	15.7	19.4	15.4	19.4	15.6	20.1
Pelleted grain mix						
Soybean meal, 48% CP	—	—	13.8	41.9	6.92	22.6
Fishmeal, menhaden	4.04	16.1	—	—	2.02	10.4
Bloodmeal, spray-dried	4.04	22.4	—	—	2.02	10.4
Corn, ground	25.1	14.6	18.5	10.9	21.8	12.0
Wheat, ground	23.5	19.4	20.8	17.4	22.1	14.9
Barley, ground	—	—	4.94	3.59	2.47	1.82
Soybean hulls	2.48	1.97	0.79	0.64	1.64	1.82
Tallow	0.57	—	1.04	—	0.80	—
Sodium bicarbonate	0.45	—	0.44	—	0.45	—
Minerals and vitamins	1.87	—	2.35	—	2.10	—
Forage:concentrate ratio		37:63		38:62		38:62

^aFed to inoculum donor cow.^bBased on original ration formulation.

lized by NSC-degrading bacteria, limiting NH₃ for fiber-grading bacteria. Understanding how to maintain adequate concentrations of both ruminal NH₃ and peptide N to maximize microbial growth would allow optimal formulation of dietary RDP and RUP concentrations. Our hypothesis was that high dietary RDP would provide peptide N to optimize microbial growth and maintain NH₃ levels above 5 mg/dL, but low RDP plus urea would limit peptide N, and lower RDP without urea would limit both. Our objectives were to ascertain effects of peptide or NH₃ limitations on microbial growth, fiber digestibility, and fermentation characteristics in continuous cultures either infused or not infused with urea at physiological levels and fed low vs high RDP (8 vs 11% of DM).

Materials and Methods

Continuous Culture System

A dual-flow continuous culture apparatus (Griswold et al., 1996) with four fermentor units was used for this study. Temperature was maintained at 39°C, and pH was controlled at 6.1 ± 0.1 using an automated peristaltic pump system with 5 N HCl or 5 N NaOH. The system was continually flushed with N₂ gas to maintain anaerobic conditions.

Inoculum was collected from a ruminally fistulated, nonlactating Holstein cow fed a basal diet consisting of grass hay, alfalfa hay, and a grain mix (Table 1) at approximately 1% of BW. The surgery protocol and animal handling procedures were approved by The Ohio State University Animal Care and Use Committee (Approval number 94-AG027). Whole ruminal contents were collected 4 h postprandially and squeezed through one layer of cheesecloth. The strained ruminal fluid (1,400 mL per fermentor) was then used to inoculate the fer-

mentor system. To provide additional particle-associated bacteria, approximately 1 kg of squeezed ruminal contents was mixed and blended with 1,500 mL of pre-warmed buffer (Weller and Pilgrim, 1974) with urea omitted. The mixture was then squeezed through one layer of cheesecloth, and the resulting buffer/ruminal fluid (300 mL per fermentor) was added to the inoculum in each fermentor.

The average fermentor volume was $1,654 \pm 40$ mL. The liquid dilution rate was 0.12 h^{-1} using the buffer of Weller and Pilgrim (1974) with or without urea. The solids dilution rate was 0.055 h^{-1} , which produced a mean solids retention time of 18 h.

Treatments and Diets Fed

Two levels of RDP (8 vs 11% of dietary DM) with or without 0.4 g/L urea provided in the artificial saliva were examined. Treatments were: 1) low RDP (8%) with urea (**LDU**), 2) high RDP (11%) with urea (**HDU**), 3) low RDP (8%) with no urea (**LDNU**), and 4) high RDP (11%) with no urea (**HDNU**).

Diet composition and nutrient analysis are provided in Tables 1 and 2, respectively. The diets consisted of an alfalfa and grass hay mixture with either a low- or high-RDP pelleted grain mix. The alfalfa and grass hays were ground through a 2-mm screen in a Wiley mill prior to mixing and feeding. Diets (100 g DM/d) were fed continually and were formulated to meet NRC recommendations for an early-lactation, Holstein, multiparous cow (NRC, 1989). All treatments were formulated to have 16, 28, and 40% ADF, NDF, and nonfiber carbohydrates (**NFC**), respectively.

Sampling and Analyses

Experimental periods were 10 d, with the initial 7 d for equilibration and the last 3 d for sampling. During

Table 2. Nutrient composition of forages, grain mixtures, and complete diets fed to continuous culture fermentors

Nutrient	Forages		Grain Mixes		Diets	
	Grass hay	Alfalfa hay	Low RDP	High RDP	Low RDP	High RDP
DM, %	93.7	84.8	87.2	87.3	91.7	91.7
			% of DM			
OM	95.6	87.8	94.7	94.2	94.4	94.6
N	0.77	3.42	3.15	2.96	2.72	2.76
CP	4.8	21.4	19.7	18.5	17.2	17.0
Rumen-degradable protein (RDP) ^a	1.67	12.54	9.06	11.79	8.35	10.94
Rumen-undegradable protein (RUP) ^b	3.14	8.86	10.65	6.73	8.89	6.05
NDF	68.3	33.4	13.1	11.7	29.9	29.8
NDIN ^c	0.28	0.82	0.32	0.15	0.38	0.26
ADF	43.2	26.4	5.4	5.4	17.1	17.1
ADIN ^d	0.13	0.23	0.05	0.05	0.11	0.12
NSC ^e	5.7	2.9	55.1	52.7	35.0	34.2
NFC ^f	22.1	36.6	59.7	60.6	46.5	45.8
Fat ^g	2.09	1.57	4.12	4.23	3.30	3.42
In situ DM degradability, % ^a	33.9	51.6	63.2	72.2	50.8	57.8
In situ N degradability, % ^a	34.8	58.2	46.0	63.7	48.4	64.4

^aDetermined by in situ degradability procedures of Callison et al. (2001).

^bDetermined by the formula: RUP = CP – RDP.

^cNeutral detergent-insoluble nitrogen.

^dAcid detergent-insoluble nitrogen.

^eNonstructural carbohydrates (NSC) determined by enzymatic analysis.

^fNonfiber carbohydrates (NFC) determined by difference.

^gDetermined by the formula: Fat = LCFA/0.9. Long chain fatty acids (LCFA) were measured by the procedures of Pantoja et al. (1994).

sampling periods, effluent was collected in containers held at 4°C by a coldwater bath to limit bacterial fermentation. Total effluent from each 24 h within the 3-d sampling period was mixed, and a 1-L subsample was composited each day and stored at 4°C for later analysis. Effluent composites were lyophilized and then ground through a 1-mm screen in a Cyclotec Sample Mill (Foss North America, Eden Prairie, MN).

On d 10, fermentor contents were allowed to settle, and the fluid layer was collected by vacuum through a 150-mesh screen. The fluid layer was centrifuged (500 × g, 20 min, 4°C) to remove any protozoa and feed particles, and the resulting supernate was centrifuged (20,000 × g, 15 min, 4°C) to pellet bacteria. The bacterial pellet was resuspended and recentrifuged, first with saline (8.5 g NaCl/L) and repeated using 50% methanol. The final pellet was resuspended in distilled H₂O, lyophilized, and stored for later analysis.

Soluble N fractions in the fermentor effluent were determined in the following manner. Subsamples (30 mL) were collected from homogenized 24-h effluent composites into clean 40-mL centrifuge tubes using individual, clean, wide-bore pipettes on d 8, 9, and 10 of each experimental period. Samples were centrifuged immediately after collection (15,000 × g, 20 min, 4°C), and the supernate was decanted and filtered through Whatman #1 filter paper into a clean 30-mL bottle for storage at –20°C until analysis. Soluble protein-N in the filtered samples was determined using the Lowry protein assay (Lowry et al., 1951) with BSA as the standard, and protein values divided by 6.25. Samples for trichloroacetic acid (TCA)-soluble N analysis were prepared by acidifying filtered supernate (20 mL) using 50% (wt/vol) TCA to a final

TCA concentration of 10% (vol/vol). The acidified samples were stored at 4°C for 24 h and then centrifuged (20,000 × g, 15 min, 4°C) to pellet any TCA-precipitable material. Supernates were decanted and stored at –20°C until analyzed for N using the microKjeldahl procedures of Oldick et al. (2000). The TCA-soluble N fraction was expected to contain peptide, AA, and NH₃ nitrogen. Ammonia N in the filtered samples was measured using the colorimetric assay of Chaney and Marbach (1962), and the NH₃ value multiplied by 0.8235 to convert to N.

Concentrations of ADF and NDF in feed and effluent were determined by the methods of Van Soest et al. (1991). Feed, bacteria, and effluent samples were analyzed for DM, ash, OM, and N according to methods of the AOAC (1990). Bacterial and effluent purine concentrations were determined by the procedure of Zinn and Owens (1986) as modified by Ushida et al. (1985). Long chain fatty acids (LCFA) in feed and effluent and VFA in effluent were determined according to the methods of Pantoja et al. (1994). Dietary and effluent NSC were determined using the hydrolysis method of Casper et al. (1990) and sugar quantification method of Blakeney et al. (1983).

Ruminal DM and CP degradation of dietary components and diets were determined by in situ procedures using Dacron bags. The in situ procedures were performed after the donor cow had been on the inoculum diet for 21 d but before the start of the continuous culture experiment. Samples were ground through a 1-mm screen in a Wiley mill, and were weighed (1 g) into Dacron bags (pore size = 50 ± 10 µ; Ankom Co., Fairport, NY). Bags were sealed, and duplicate bags were placed in the rumen after a 15-min preincubation in water start-

Table 3. Treatment least squares means for the effects of rumen-degradable protein (RDP) concentration and urea addition on N partitioning and bacterial efficiency in continuous culture

Partitioning	Treatment ^a				SEM	Contrast ^b		
	LDNU	HDNU	LDU	HDU		Urea	RDP	Urea × RDP
Dietary N intake, g/d	2.72	2.76	2.72	2.76				
CP degraded, %	38.4	51.9	56.8	69.3	7.76	0.005	0.026	NS
NH ₃ nitrogen, mg/dL ^c	0.26	0.21	6.06	6.11	0.43	<0.001	NS	NS
TCA-soluble N, mg/dL ^d	2.24	2.62	6.55	6.75	1.41	0.002	NS	NS
Peptide N, mg/dL ^e	1.99	2.41	0.49	0.63	1.24	0.061	NS	NS
Soluble-protein N, mg/dL ^f	2.43	4.11	3.69	4.54	0.52	0.013	0.008	NS
Total N flow, g/d	2.81	2.86	3.39	3.43	0.09	<0.001	NS	NS
NAN, g/d ^g	2.80	2.85	3.11	3.14	0.11	0.003	NS	NS
NANMN, g/d ^h	1.70	1.31	1.19	0.83	0.21	0.005	0.02	NS
Microbial N								
g/d	1.10	1.54	1.92	2.31	0.19	<0.001	0.01	NS
g/kg of OM truly digested	19.9	24.9	28.0	32.2	2.58	0.002	0.021	NS
g/kg of CHO digested ⁱ	24.1	31.3	35.5	44.3	3.93	0.002	0.012	NS
N capture, % ^j	39.6	56.4	52.7	64.0	7.10	0.012	0.016	NS
N efficiency, % ^k	39.7	56.7	69.4	84.8	6.88	<0.001	0.007	NS

^aLDNU = low RDP (8% of dietary DM) with no urea, HDNU = high RDP (11% of dietary DM) with no urea, LDU = low RDP (8% of dietary DM) with urea, and HDU = high RDP (11% of dietary DM) with urea.

^bProbability of main effects of urea (0 vs 0.4 g/L of artificial saliva), RDP (8 vs 11% of DM), or the urea × RDP interaction. NS = ($P > 0.10$).

^cDetermined by the colorimetric method of Chaney and Marbach (1962) with NH₃ nitrogen value multiplied by 0.8235.

^dNitrogen soluble in 10% (vol/vol) trichloroacetic acid; determined by the microKjeldahl procedure of Oldick et al. (2000).

^eDetermined by the formula: peptide N = trichloroacetic acid-soluble N – NH₃ N.

^fDetermined by the procedure of Lowry et al. (1951) with protein value divided by 6.25.

^gNonammonia N (NAN) determined by the formula: NAN = total N flow – NH₃ N.

^hNonammonia, nonmicrobial N (NANMN) determined by the formula: NANMN = NAN – microbial N.

ⁱCHO = carbohydrate.

^jPercentage of total N intake (dietary and saliva) converted to microbial N.

^kPercentage of dietary N intake converted to microbial N.

ing at 0800. The ruminal incubations were performed twice during two consecutive weeks for a total of four replications per sample per time point. The bags were placed in reverse order to achieve the following incubation times: 0, 1, 3, 6, 9, 12, 24, 36, 48, 60, 72, and 96 h. At 96 h, all bags were removed and rinsed thoroughly in cold tap water until no coloration of the water occurred. The bags were then dried at 55°C, and DM and N determined (not corrected for bacterial contamination). Samples were prepared and analyzed for N using the microKjeldahl procedures of Oldick et al. (2000). The rate of disappearance of potentially digestible N was determined by the log-linear procedure to provide initial estimates for the nonlinear procedure of SAS (SAS Inst., Inc., Cary, NC), as described by Callison et al. (2001).

Calculations

Dietary and effluent NFC concentration was calculated using the formula: $\text{NFC} = 100 - [\text{ash} + \text{CP} + (\text{NDF} - \text{NDICP}) + \text{fat}]$, where neutral detergent-insoluble CP (NDICP) was determined by multiplying neutral detergent-insoluble N by 6.25, and fat was estimated as $\text{LCFA}/0.9$, as triglycerides were assumed to be 90% fatty acids. Hemicellulose concentration was calculated as $\text{NDF} - \text{ADF}$. Peptide and AA nitrogen (termed peptide

N) was calculated by subtracting NH₃ nitrogen from TCA-soluble N.

Experimental Design and Statistical Analysis

The experimental design was a 4×4 Latin square with a 2×2 factorial arrangement of treatments. Data were statistically analyzed using the PROC GLM procedures of PC SAS (SAS Inst., Inc.). The statistical model included the fixed effects of period, fermentor, urea, RDP, and the interaction of urea by RDP. The model as fitted was:

$$Y_{ijkl} = \mu + F_i + P_j + U_k + R_l + UR_{kl} + \varepsilon_{ijkl}$$

where Y_{ijkl} is the dependent variable, μ is the overall mean, F_i is the fixed effect of the i th fermentor ($i = 1, \dots, 4$), P_j is the fixed effect of the j th period ($j = 1, \dots, 4$), U_k is the fixed effect of the k th urea level ($k = 1, 2$), R_l is the fixed effect of the l th RDP level ($l = 1, 2$), UR_{kl} is the fixed interaction effect of the k th urea level by l th RDP level, and ε_{ijkl} is the residual error. One data cell was excluded from the data set due to a fermentor malfunction during the second experimental period, which resulted in the collection of only one 24-h effluent composite. Significance was determined at $P \leq 0.05$.

Table 4. Treatment least squares means for the effects of rumen-degradable protein (RDP) concentration and urea addition on digestibility of carbohydrate fractions in continuous culture

Dietary component	Treatment ^a				SEM	Contrast ^b		
	LDNU	HDNU	LDU	HDU		Urea	RDP	Urea × RDP
DM	42.7	47.9	57.7	60.4	1.91	<0.001	0.014	NS
OM	30.1	39.9	45.7	50.1	3.38	<0.001	0.014	NS
NDF	47.5	53.2	52.7	52.5	1.66	0.02	0.046	0.022
ADF	38.7	41.1	46.0	46.8	3.08	0.009	NS	NS
Hemicellulose ^c	55.8	70.6	61.6	60.2	5.62	NS	NS	0.045
NSC ^d	73.6	83.4	92.5	93.0	1.50	<0.001	0.004	0.003
NFC ^e	50.8	56.4	62.4	63.2	1.20	0.003	NS	NS

^aLDNU = low RDP (8% of dietary DM) with no urea, HDNU = high RDP (11% of dietary DM) with no urea, LDU = low RDP (8% of dietary DM) with urea, and HDU = high RDP (11% of dietary DM) with urea.

^bProbability of main effects of Urea (0 vs 0.4 g/L of artificial saliva), RDP (8 vs 11% of DM), or the urea × RDP interaction. NS = ($P > 0.10$).

^cDetermined as NDF – ADF.

^dNonstructural carbohydrates (NSC) determined by enzymatic analysis.

^eNonfiber carbohydrates (NFC) determined by difference.

Results

In Situ Degradability of Ingredients and Diets

The in situ degradabilities of the forages, pelleted grain mixtures, and the experimental diets are provided in Table 2. The experimental diets were formulated to provide 8% (low degradable) and 11% (high degradable) of the diet DM as RDP. The in situ CP degradabilities of the low- and high-RDP diets were 48.4 and 64.4%, respectively. These in situ CP degradabilities were calculated to provide 8.35 and 10.94% of dietary DM as RDP for the low- and high-RDP diets, respectively (Table 2).

Degradability of Crude Protein, Nitrogen Partitioning, and Microbial Efficiency

No interactions between urea infusion and RDP concentration were detected for degradability of CP, N partitioning, or microbial efficiency (Table 3). Degradability of CP in continuous culture was increased by urea infusion ($P = 0.005$) and increased dietary RDP ($P = 0.026$). The CP degradability of the diets determined by in situ procedures (Table 2) was proportionally similar to the CP degradation values for the low- and high-RDP diets in continuous culture (Table 3). This similarity in CP degradabilities suggested that the experimental objective to test the effect of RDP concentration on microbial growth and fermentation was achieved.

Among soluble N fractions, NH_3 nitrogen, TCA-soluble N, and soluble-protein N were all significantly increased by infusion of urea. There was a trend ($P = 0.061$) for peptide N in the effluent to decrease with the infusion of urea to the fermentor. Soluble-protein N was significantly increased when RDP was 11% of the diet DM compared to when RDP was 8% of the diet DM.

Flows of total N and nonammonia N (NAN) were significantly increased and flow of nonammonia, nonmicro-

bial N (NANMN) was significantly decreased when urea was provided in the artificial saliva. The amount of NANMN flowing per day was significantly less when RDP was 11% of the diet DM compared to when RDP was 8% of diet DM. Microbial N flow (g/24 h) was significantly increased with urea infusion and with increased RDP level.

Microbial efficiency, regardless of the method of expression, showed the same main effects of urea infusion and increased RDP concentration, with no interaction detected. Provision of urea to treatments significantly improved microbial efficiency compared to treatments without urea. Greater concentrations of RDP in the diet (11% of DM) significantly increased microbial efficiency compared to diets containing less RDP (8% of DM). Both N capture and N efficiency were significantly improved with the infusion of urea compared to treatments without urea infusion. Increased concentrations of RDP in the diets significantly improved N capture and N efficiency compared to diets with less RDP.

Digestibility of Dry Matter, Organic Matter, and Carbohydrate Fractions

The effects of urea and RDP levels on digestibility of DM, OM, NDF, ADF, hemicellulose, NSC, and NFC are shown in Table 4. There were specific urea × RDP interactions, as evidenced by the significantly decreased digestibility of NDF, hemicellulose, and NSC for LDNU; digestibility of hemicellulose was significantly greater, but that of NSC significantly less, for HDNU than LDU or HDU.

Urea (0.4 g/L) in the artificial saliva significantly increased digestibility of DM, OM, ADF, and NFC. Increasing the level of RDP in the diet from 8 to 11% of DM significantly increased digestibility of DM and OM.

Volatile Fatty Acid Concentrations

The effects of urea and RDP level on total VFA concentration and molar percentages of individual VFA are

Table 5. Treatment least squares means for the effects of rumen-degradable protein (RDP) concentration and urea addition on VFA concentrations in continuous culture

Parameter	Treatment ^a				SEM	Contrast ^b		
	LDNU	HDNU	LDU	HDU		Urea	RDP	Urea × RDP
Total VFA, mM	73.0	80.5	97.5	100.1	8.32	0.003	NS	NS
Individual VFA, mol/100 mol								
Acetate	53.7	56.0	56.3	60.2	3.15	0.08	NS	NS
Propionate	27.1	25.1	26.1	24.7	3.91	NS	NS	NS
Butyrate	14.1	13.6	13.9	11.7	2.49	NS	NS	NS
Isobutyrate	0.63	0.35	0.48	0.30	0.15	NS	0.047	NS
Isovalerate	1.10	0.99	1.40	1.03	0.31	NS	NS	NS
Valerate	3.33	3.84	1.85	2.05	0.93	0.025	NS	NS

^aLDNU = low RDP (8% of dietary DM) with no urea, HDNU = high RDP (11% of dietary DM) with no urea, LDU = low RDP (8% of dietary DM) with urea, and HDU = high RDP (11% of dietary DM) with urea.

^bProbability of main effects of urea (0 vs 0.4 g/L of artificial saliva), RDP (8 vs 11% of DM), or the urea × RDP interaction. NS = ($P > 0.10$).

shown in Table 5. Total VFA concentration and the individual molar percentage of valerate were significantly increased as the result of urea infusion in the artificial saliva. There was a trend ($P = 0.08$) for urea infusion to increase acetate molar percentage. Diets with greater amounts of RDP (11% of diet DM) significantly increased the molar percentage of isobutyrate compared to diets with less RDP (8% of diet DM).

Discussion

To meet the objectives of this study, urea infusion levels and RDP concentration were set sufficiently apart to elicit a measurable response with the baseline treatment, LDNU, designed to be the most N-limiting (i.e., negative control). In vivo studies examining the effects of urea and RDP have often produced mixed results, which were presumably due to availability of NH_3 released from urea recycling through the rumen wall and the saliva. Therefore, to determine if urea recycling would mask potential RDP effects, either urea was either provided or not provided in the artificial saliva. The level of urea infused was based on physiological levels found in ruminant saliva (Weller and Pilgrim, 1974), and provision of urea in the artificial saliva increased potential total available N in the treatments approximately 35 to 44%. Additionally, Griswold et al. (1996) indicated that diets with 10% of DM as RDP and low protein degradability (<50%) would limit N available for digestibility, fermentation, and microbial growth in continuous culture. Low-RDP treatments provided RDP at 8.35% of diet DM. High-RDP treatments provided RDP at 10.94% of diet DM or approximately 65% of the CP content, which was representative of current feeding standards for dairy cattle (NRC, 1989). Considering the degradation of dietary CP (Table 3), available N for the LDNU, HDNU, LDU, and HDU treatments was 1.04, 1.43, 1.54, and 1.91% of dietary DM, respectively. Nitrogen was then limiting for treatments LDNU and HDNU according to previous research (Griswold et al., 1996).

Nitrogen Fractionation and Microbial Protein Synthesis

Urea infusion significantly affected all of the N partitioning and microbial protein synthesis measurements (Table 3), which provided strong indications that N was limited with the LDNU and HDNU treatments. The increase in CP degraded with urea infusion suggested that protein degradation was dependent on the size or mass of the microbial population, which, in turn, was dependent on available N. This finding agreed with Windschitl and Stern (1988), who found that protein degradation responded in a quadratic manner to urea infusion in continuous culture, with the greatest response at 0.4 g urea/L of artificial saliva. Milton et al. (1997) reported a trend for ruminal N digestibility to increase linearly with linear urea addition to N-limited finishing diets for steers. Cameron et al. (1991) also found that urea supplementation of fishmeal diets increased degradation of dietary CP in lactating cows.

The increased N digestion with urea infusion caused elevations in NH_3 nitrogen, TCA-soluble N urea, and soluble-protein N, but urea decreased peptide N in continuous culture (Table 3). Under conditions of N limitation, both the rates of protein degradation and peptide uptake appeared to be limiting for microbial growth. Increasing the concentration of dietary RDP increased CP degradation and production of soluble-protein N, but RDP did not affect NH_3 nitrogen, TCA-soluble N, or peptide N (Table 3). Chen et al. (1987) indicated that AA sequence of the peptides would control peptide uptake. The results of the current study suggested that peptide uptake was dependent on total available N in the system rather than dietary protein composition, and factors other than AA sequence may regulate peptide uptake in the rumen. Our findings agreed with Williams and Cockburn (1991), who reported low correlations between protein degradability or solubility and ruminal peptide N concentrations.

Calsamiglia et al. (1995) employed the ratio of NAN flow relative to dietary N intake to determine N usage from urea in artificial saliva and suggested that urea N

only supported 7% of the NAN flow when low RDP diets were fed in continuous culture. In the current study, the ratio of NAN:dietary N intake was 1.13 and 1.16 for the LDU and HDU treatments, respectively. However, considering the amount of degraded dietary N and the actual urea N supplied to the fermentors per day, 43.2 and 44.9% of urea N was utilized for microbial protein synthesis with the LDU and HDU treatments, respectively. Urea N, therefore, supported 19.8 and 17.3% of the microbial protein synthesis for the LDU and HDU treatments, respectively. Windschitl and Stern (1988) reported a linear response in microbial protein synthesis as urea infusion increased from 0 to 30% of the total N intake in continuous culture.

Our hypothesis, that diets high in RDP would provide enough available N to maintain NH_3 concentrations above 5 mg/dL while also providing enough peptides and AA to optimize digestibility, fermentation, and microbial growth when urea is not available as a source of N, was incorrect. Our hypothesis had assumed that peptide and AA nitrogen were the limiting N forms for microbial growth under all conditions. Changes in the concentrations of N fractions could be used to determine the most limiting form of N for microbial growth under varying conditions. Under N-limiting conditions without urea infusion (i.e., LDNU and HDNU), NH_3 nitrogen was extremely low when peptide N accumulated, which suggested that NH_3 nitrogen was the most limiting N form. When urea was infused, which provided ample N (i.e., LDU and HDU), NH_3 nitrogen accumulated, but peptide N concentrations became extremely low, which suggested that peptide N was the most limiting N form for microbial growth.

Dry Matter, Organic Matter, and Carbohydrate Digestibility

Ruminally available N in the form of peptides and AA can improve the digestion of both SC and NSC by the microbial population, as evidenced by the interaction of urea and RDP in the current study. When urea was not available and RDP in the diet was increased (HDNU), NDF, hemicellulose, and NSC digestibility were significantly improved (Table 4). Windschitl and Stern (1988) found no interactions between urea infusion and dietary RDP on digestibility of carbohydrates in continuous culture. Most in vitro work has shown no response in NDF or hemicellulose digestibility when ruminally available AA nitrogen was increased (Griswold et al., 1996), and hemicellulose-degrading bacteria are suggested to be slow-growing bacteria that require all of their N to come from NH_3 (Russell et al., 1992). However, there is evidence in the literature that some hemicellulolytic bacteria (e.g., *Butyrivibrio fibrisolvens*) can utilize peptide or AA nitrogen directly for microbial protein synthesis (Cotta and Hespell, 1986). The results of the current study suggested that hemicellulose-degrading bacteria might have the ability to effectively compete with NSC-degrading bacteria for peptide and AA nitrogen when

N is limiting, and therefore, support a population size sufficient to effectively degrade dietary hemicellulose.

Not all ruminal bacteria can utilize NH_3 nitrogen, but NH_3 nitrogen is the sole N source for cellulose-degrading bacteria in the rumen (Russell et al., 1992). Windschitl and Stern (1988) indicated that SBM diets could maintain NH_3 nitrogen above 5 mg/dL, which is the concentration recommended for optimal microbial growth in vitro (Satter and Slyter, 1974), without urea infusion in continuous culture. In the current study, NH_3 nitrogen was only greater than 5 mg/dL when urea was infused (Table 3). Consequently, urea infusion increased NH_3 nitrogen availability, and thereby, digestibility of carbohydrates (Table 4). These findings, along with the elevated valerate levels found with no urea infusion (Table 5), suggested that AA released from RDP were being deaminated to produce NH_3 rather than being directly incorporated into bacteria. Because ADF digestibility was not improved, whereas NSC digestibility was improved with high-RDP diets compared to low-RDP diets (Table 4), cellulolytic bacteria appeared unable to compete with NSC-degrading bacteria for NH_3 nitrogen released by AA deamination. Several studies have shown an increase in ADF or cellulose degradation when ruminal digestibility of AA nitrogen sources increased in vitro (Windschitl and Stern, 1988; Griswold et al., 1996). However, using lactating dairy cattle, McCarthy et al. (1989) found no effect of changing dietary RDP level on the ruminal digestion of SC, perhaps because of extensive urea recycling or protozoal-mediated proteolysis or deamination.

Microbial Fermentation

The decrease in the molar percentage of isobutyrate when dietary RDP increased (Table 4) was contrary to the findings of Windschitl and Stern (1988) and Calsamiglia et al. (1995), who found no difference in isobutyrate concentration when SBM, lignosulfonate-treated SBM, fishmeal, or bloodmeal were the major N sources in continuous culture. Supplementation of peptide N, up to 30% of total dietary N, linearly increased isobutyrate concentration in continuous culture (Jones et al., 1998). Isobutyrate is a fermentation endproduct of valine catabolism in the rumen (Blackburn, 1965), which suggests that valine catabolism was greater with the low-RDP diet compared to the high-RDP diet. Estimated dietary valine levels were 5.51 and 4.65% of CP for the low- and high-RDP diets, respectively, which may have contributed to the greater isobutyrate concentration with the low-RDP diets. However, when Griswold et al. (1996) fermented diets with the same AA profile that differed in the form of available N, whole protein, peptides, or AA, isobutyrate concentration increased when peptides or AA replaced whole protein, suggesting that rates of protein degradation might also affect AA catabolism and formation of branched chain VFA.

The significance of valerate accumulation on microbial growth and fermentation under the N-limiting condi-

tions of the current study was unclear. Cline et al. (1958) reported that either valeric acid synthesis increased or utilization decreased when N was limited in batch cultures of mixed rumen microorganisms. Windschitl and Stern (1988) indicated that urea infusion lowered valerate concentrations in continuous culture when SBM or lignosulfonate-treated SBM diets were fermented. However, valerate, which is an endproduct of ruminal carbohydrate and AA fermentation, has been shown to have little to no effect on microbial protein synthesis (Russell and Sniffen, 1983) or digestibility of structural carbohydrates (Gorosito et al., 1985). Valerate is one of the fermentation endproducts of lysine and arginine catabolism (Blackburn, 1965), and both of these AA contain two amino groups. These AA could have been preferentially deaminated when NH_3 nitrogen was limited because of their greater potential to generate NH_3 nitrogen.

Implications

Five milligrams per deciliter of ruminal NH_3 has been considered necessary for adequate fiber digestibility. However, hemicellulolytic bacteria might be able to effectively compete with nonstructural carbohydrate-digesting bacteria for AA nitrogen in the rumen, and ruminal NH_3 may only be necessary to support cellulolytic bacteria. The amount and form of ruminally degradable protein might need to be changed in dairy rations containing forage sources differing proportion of cellulose and hemicellulose to optimize ruminally degradable protein supply while minimizing urinary N loss into the environment.

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